# Assignment 3 — Reinforcement Learning

## 1. SupplyRun: Off-Policy Evaluation and Offline Learning

### 1.1 Setup and implementation

The supplied SupplyRun artifact defines a finite MDP with 1,920 non-terminal states, four actions (UP, DOWN, LEFT, RIGHT), discount factor $\gamma = 0.99$, and a 200-step trajectory cap. The pipeline generated 2,000 trajectories using only the supplied stochastic behaviour policy $b(a|s)$. Each trajectory stores states, actions, rewards, behaviour-policy action probabilities, next-state indices, termination flags, and a valid-step mask. Importance ratios are computed in log space to avoid numerical underflow/overflow.

Offline control replays stored trajectories without additional environment interaction. Q-learning and Expected SARSA were implemented, with Expected SARSA investigated at $\varepsilon_\pi \in \{0.05, 0.1, 0.2, 0.3\}$. A one-step importance-sampled SARSA variant was also tested, with a clipped variant included for stability analysis. The main reported configuration uses 800 replayed trajectories per seed, $\alpha = 0.1$, and 10 learning seeds.

### 1.2 Behaviour-data statistics

| Statistic | Result |
|---|---|
| Number of trajectories | 2,000 |
| Discounted return, mean ± SD | −13.998 ± 10.086 |
| Discounted return range | [−17.375, 80.973] |
| Undiscounted return, mean ± SD | −17.940 ± 15.527 |
| Mean trajectory length | 30.653 steps |
| Terminated / truncated | 99.9% / 0.1% |
| Positive-return trajectories | 1.7% |
| Exact $V^b(s_0)$, finite-horizon DP | −13.819 |

The empirical mean discounted return differs from the exact behaviour-policy value by 0.178, smaller than the reported standard error of approximately 0.226. This is a useful consistency check for the generated dataset and return calculation.

### 1.3 Offline-control results

| Method | Final greedy discounted return |
|---|---|
| Q-learning | 79.770 ± 1.242 |
| Expected SARSA, $\varepsilon_\pi = 0.05$ | 70.409 ± 30.844 |
| Expected SARSA, $\varepsilon_\pi = 0.1$ | 70.409 ± 30.844 |
| Expected SARSA, $\varepsilon_\pi = 0.2$ | 70.409 ± 30.844 |
| Expected SARSA, $\varepsilon_\pi = 0.3$ | 50.357 ± 46.770 |
| IS-SARSA, all tested $\varepsilon_\pi$ | 11.494 ± 46.495 |

Values are mean ± standard deviation over 10 seeds. Q-learning achieved the strongest and most consistent result at the specified 800-trajectory replay budget. Expected SARSA at $\varepsilon_\pi = 0.05$, 0.1, and 0.2 converged to the same greedy outcomes across seeds, but this does not imply identical underlying policies or Q-functions. The unclipped IS-SARSA update was unstable at the selected step size; sensitivity checks reported Q-value magnitudes growing to around $10^{14}$ during extended training. Clipping ratios or reducing the step size improved stability, with clipping introducing bias.

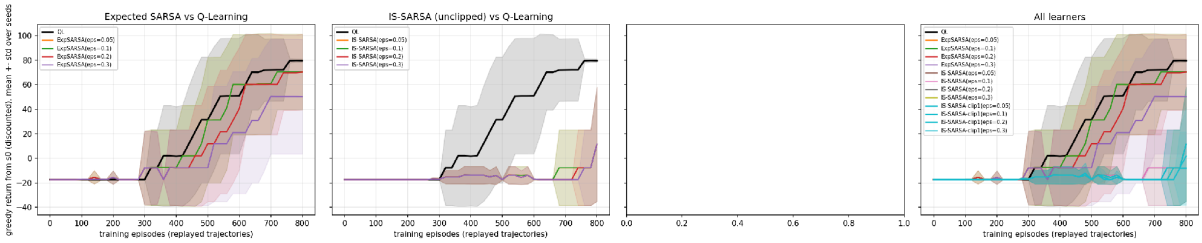



Figure 1. Offline learning curves.

## 1.4 Off-policy evaluation

For target policy π, trajectory importance ratios are $\rho_i = \prod_t \pi(A_t|S_t)/b(A_t|S_t)$. Ordinary importance sampling (OIS) averages $\rho_i G_i$ and is unbiased but can have very high variance. Weighted importance sampling (WIS) normalizes the weights, reducing variance at the cost of finite-sample bias. Per-decision IS (PDIS) applies cumulative ratios to each discounted reward.

Effective sample size was only about 0.10%–0.45% of the 2,000 trajectories across the reported target policies. Many trajectories therefore contribute almost nothing, and a small number dominate. OIS estimates can be far from finite-horizon DP reference values; WIS remains bounded by observed returns but can be biased toward the few trajectories with appreciable weights.

OIS and PDIS are identical for every reported target policy. This is consistent with the reward structure if essentially all non-zero reward occurs at the terminal transition, where the cumulative per-decision ratio equals the full trajectory ratio.

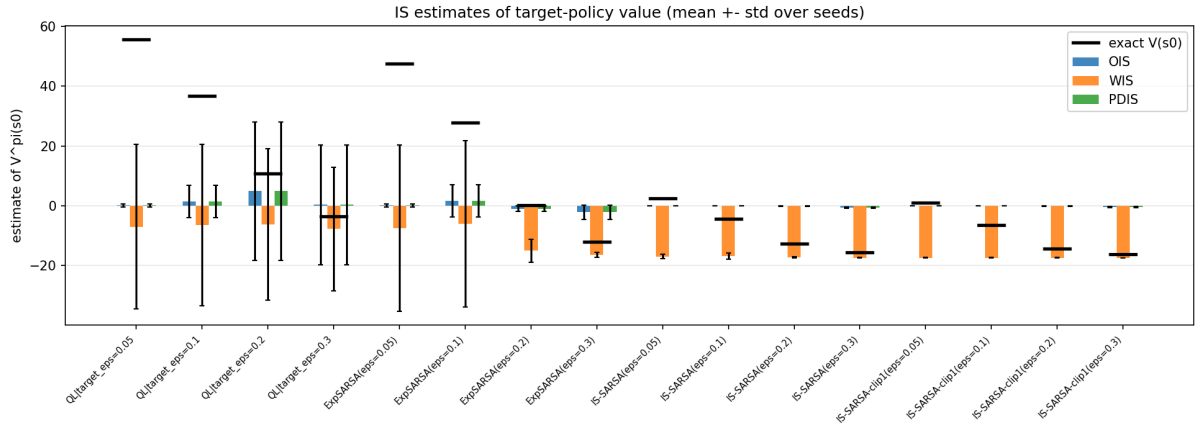



Figure 2. Importance-sampling estimates against ground truth.

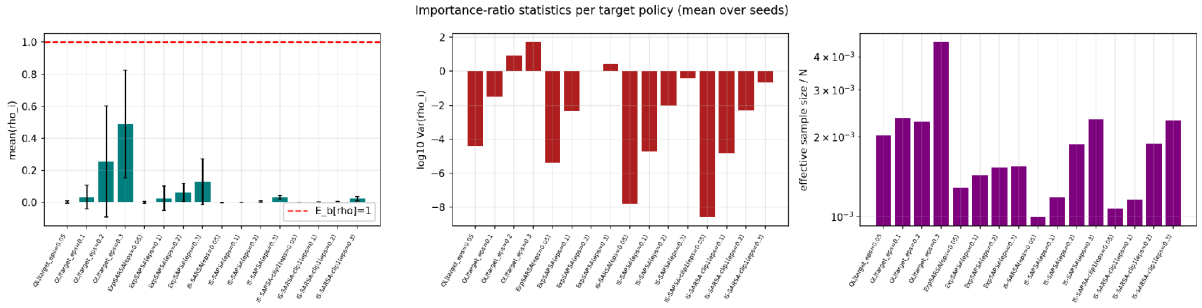



Figure 3. Importance-ratio statistics.

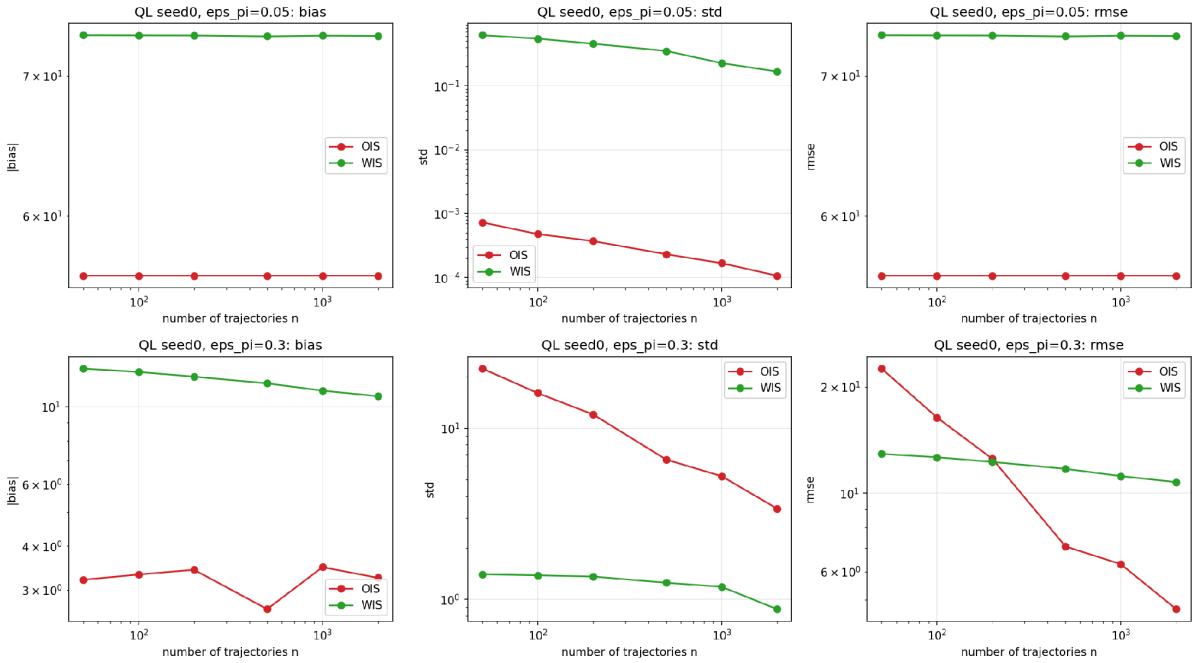



Figure 4. OIS versus WIS bias and variance.

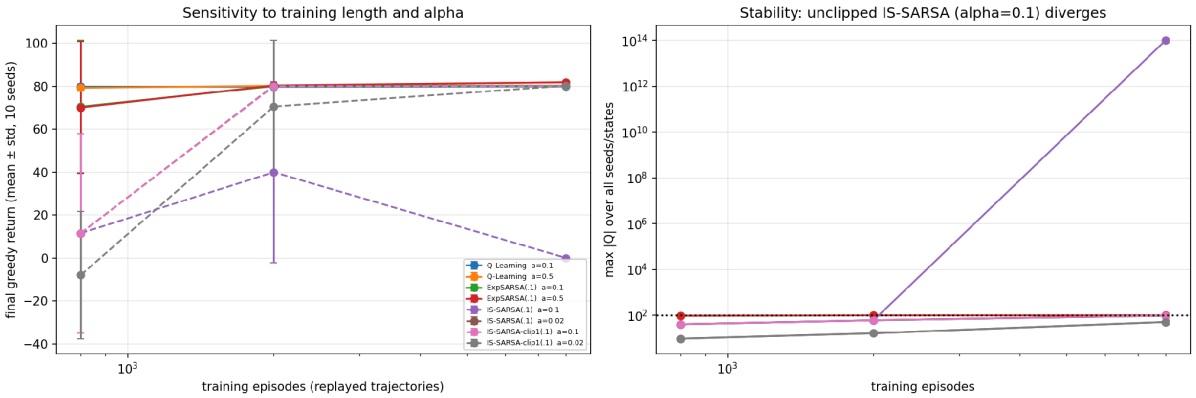



Figure 5. Sensitivity and IS-SARSA stability.

# 2. Policy Gradients on InvertedPendulum-v5

## 2.1 Environment and protocol

The experiments use Gymnasium's default InvertedPendulum-v5 environment: observation dimension 4, action dimension 1, action bounds [−3, 3], and maximum episode length 1,000. Reward is +1 per step until termination, so undiscounted return equals episode length. All experiments use γ = 1, fixed Gaussian action standard deviation σ = 0.3, batches of 16 complete episodes, Adam, and 7,000 training episodes per run. Results use seeds 0–4; final evaluation return is the mean of the last 20 evaluation episodes.

The discretized policy uses eight bins per observation dimension ($8^4$ = 4,096 table entries), with ranges: cart position [−0.5, 0.5], pole angle [−0.2, 0.2], cart velocity [−1, 1], and pole angular velocity [−2, 2]. Out-of-range values map to the first or last bin. The neural policy is an MLP 4→64→64→1 with ReLU hidden layers and linear output; its final layer is zero-initialized.

Actions are sampled as an unclipped Gaussian variable u; only the environment action is clipped to [−3, 3]. Log probability is evaluated at u, the variable actually sampled from the Gaussian.

## 2.2 Methods

| ID | Policy / estimator |
|---|---|
| **M1** | Discretized table + vanilla full-return REINFORCE |
| **M2** | Neural policy + vanilla full-return REINFORCE |
| **M3** | Neural policy + leave-one-out batch baseline |
| **M4** | Neural policy + moving-average baseline, β = 0.01 |
| **M5** | Neural policy + moving-average baseline, β = 0.1 |
| **M6** | Neural policy + return-to-go |

The table policy used learning rate 0.02; neural methods used 0.001, selected using a small pilot. Thus, the table-versus-neural comparison is not fully controlled for learning rate. The M3 leave-one-out baseline passed its numerical equivalence check against the corrected batch-mean form with maximum absolute gradient difference 0.

## 2.3 Results

| Method | Final eval return, mean ± SD | Seeds reaching H = 900 | Mean threshold episode (successful seeds) |
|---|---|---|---|
| **M1** | 147.82 ± 44.33 | 0/5 | — |
| **M2** | 150.73 ± 76.97 | 0/5 | — |
| **M3** | 1000.00 ± 0.00 | 5/5 | 3168 ± 444 |
| **M4** | 237.97 ± 355.43 | 2/5 | 4560 ± 339 |
| **M5** | 815.71 ± 353.22 | 3/5 | 3627 ± 333 |
| **M6** | 615.35 ± 507.75 | 5/5 | 4288 ± 1486 |

M3 was the most reliable method: every seed reached the threshold and finished at the maximum return. Vanilla REINFORCE (M1 and M2) did not learn a strong balancing policy under this configuration. M4's slow moving-average baseline often lagged behind the changing return scale; increasing β to 0.1 improved the mean result, though variability remained high.

M6 reached the threshold at some point in all five seeds, but its final performance was unstable: two seeds ended below 100, leaving a final mean of 615.35. A threshold crossing should therefore not be confused with sustained performance.

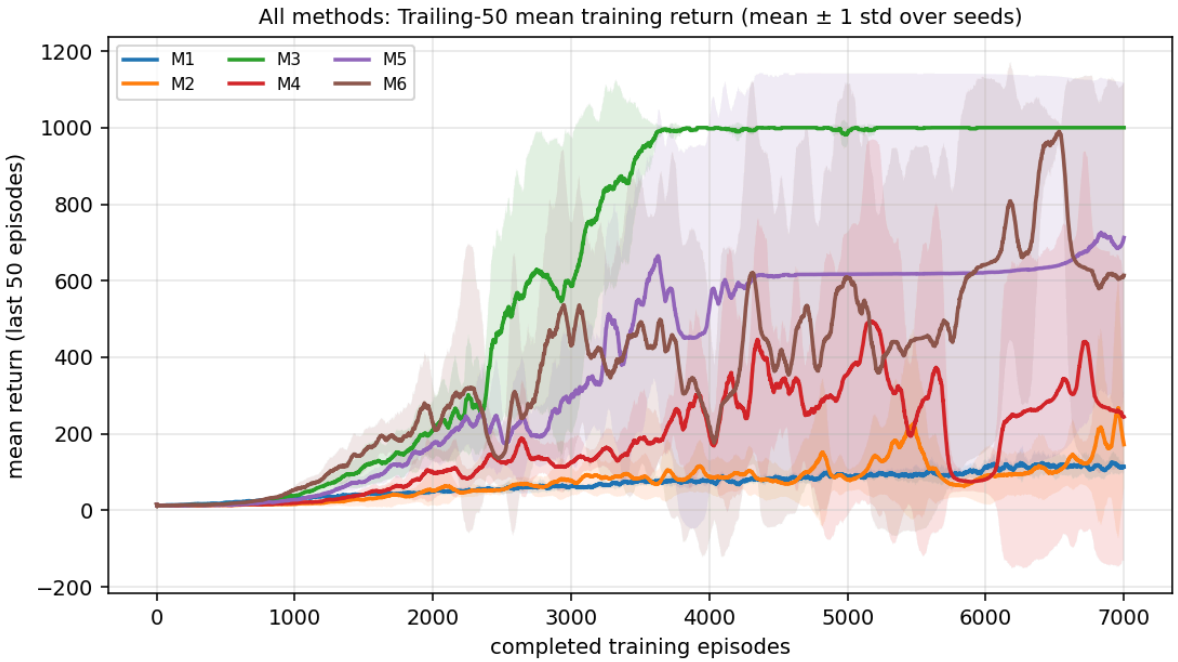



Figure 6. Smoothed learning curves across policy-gradient methods.

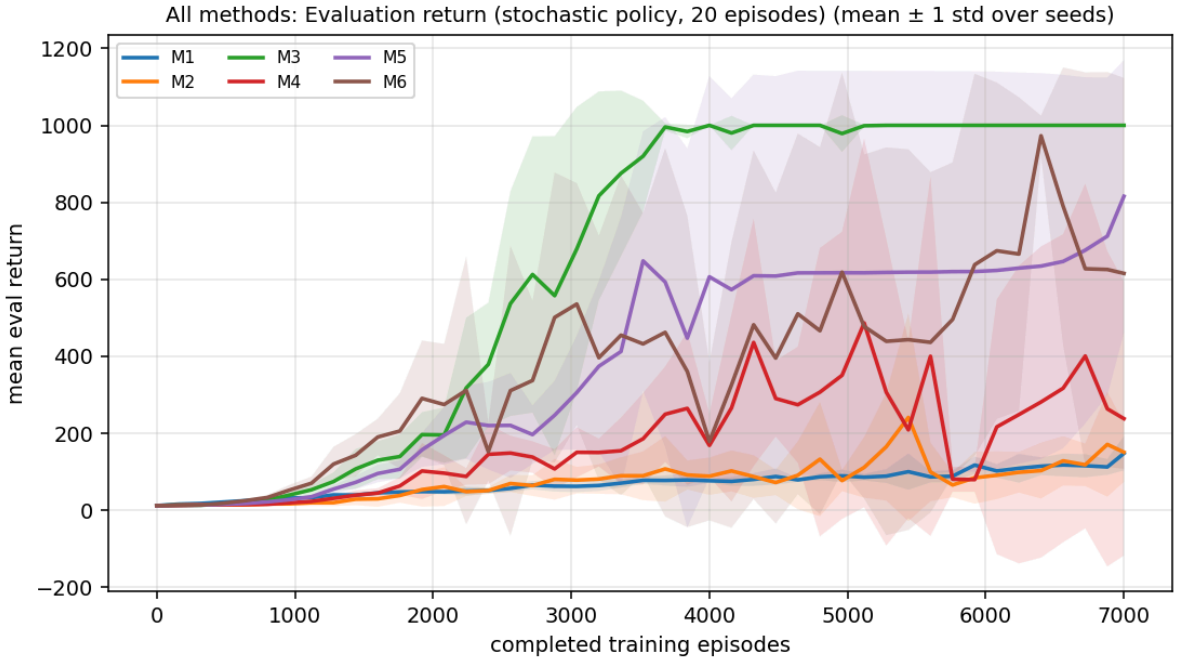



Figure 7. Evaluation return comparison.

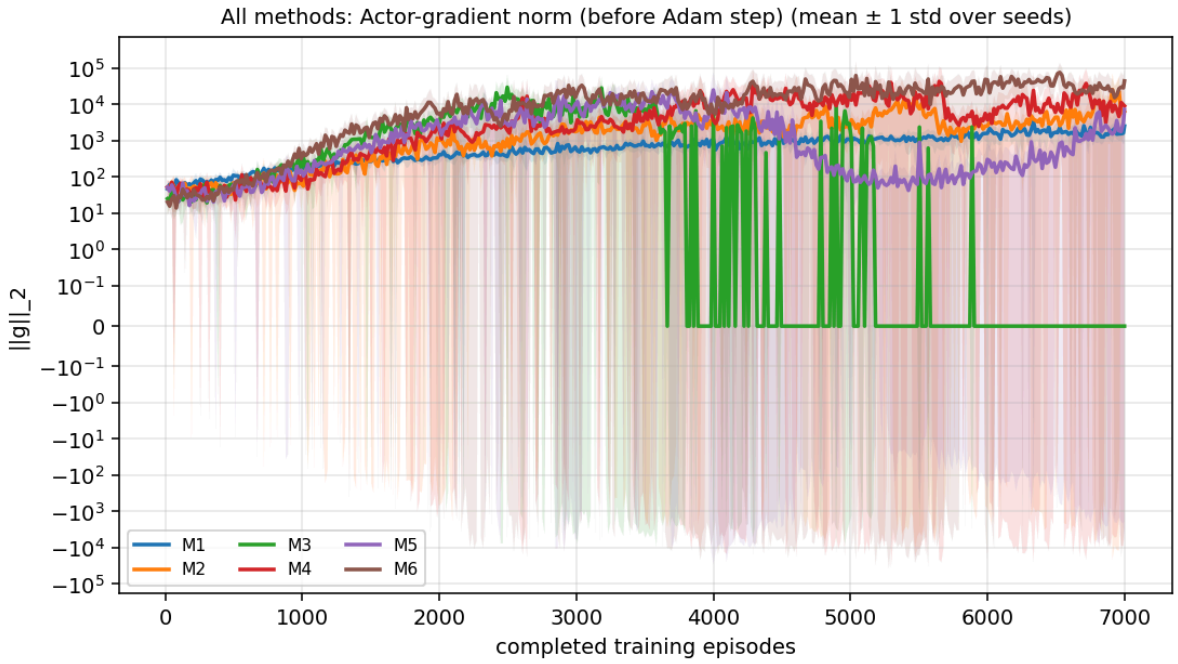



Figure 8. Gradient norm comparison.

## 2.4 Interpretation

First, baseline choice mattered: M3 substantially outperformed vanilla REINFORCE in these runs. Its leave-one-out baseline excludes episode i's return when forming the baseline, preserving the intended independence from that episode's action samples. Second, a moving baseline must track the learning process; β = 0.01 was often too slow, while β = 0.1 performed better on average. Third, return-to-go removes past rewards from the weight multiplying each timestep's score-function gradient, preserving the expected gradient while generally reducing variance, but it did not eliminate seed-level instability.

The table policy learns independently for each discretized state and does not generalize across bin boundaries. The neural policy shares parameters across states, but vanilla REINFORCE still performed poorly here. Since M1 used a different learning rate from M2–M6, the results should not be interpreted as a clean causal comparison of parameterization alone.

# 3. Sanity check and limitations

- Dataset return check passes: empirical mean discounted return −13.998 is within one standard error of exact $V^b(s_0)$ = −13.819.
- M3 gradient implementation check passes: leave-one-out and corrected batch-mean gradients match exactly in the recorded check.
- OIS = PDIS may be explained by terminal-only non-zero rewards; confirm the reward structure if challenged in the viva.
- Offline learning is data-limited: with 800 replays, outcomes depend on whether each seed sees enough successful trajectories. The main results retain the specified 800-episode setting.
- Policy-gradient runs M2 and M6 can peak and later collapse; final returns, threshold crossings, and learning curves should be interpreted together.
- The archive and saved results were inspected; no dependencies were installed and no training was rerun for this review.

## 4. Conclusion

For offline SupplyRun, Q-learning achieved the best final greedy return under the specified replay budget, while importance-sampling evaluation was severely limited by low effective sample size. For InvertedPendulum, REINFORCE with a leave-one-out batch baseline (M3) was the strongest and most stable configuration. Together, the experiments illustrate the importance of baseline design, importance-weight degeneracy, data coverage, and seed-level stability when interpreting RL results.